The Chinese University of Hong Kong

Department of Computer Science and Engineering

KTL2401 - Invariant Hand Gesture Representation Learning with Augmented Contrastive Learning

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Abstract

Hand Gesture Recognition (HGR) has played an important role in driving fields such as virtual/augmented/mixed reality (VR/AR/MR) and human-machine interactions. However, while existing deep learning approaches that utilize keypoint landmarks have achieved excellent performance in hand gesture recognition tasks [1], many existing frameworks rely heavily on fixed sets of gestures in the training dataset, limiting their ability to generalize over uncommon orientations or hand poses.

In this work, we explored a combination of semi-supervised contrastive learning approach with a large synthetically generated corpus of realistic hand configurations, in order to expose models to a significantly wider range of orientations and poses that are typically unavailable in limited real datasets. These approaches have been proven effective on datasets involving only static gestures, while the extension of the contrastive approach to dynamic gesture recognition tasks remain challenging, suggesting the need for more sophisticated methods that should be discussed in later work.

1 Introduction

1.1 Background

Hand Gesture Recognition (HGR) has played vital role in driving fields such as virtual/augmented/mixed reality (VR/AR/MR) and human-machine interactions. In particular, HGR is frequently used to extract instructions or language conveyed through hand gestures. However, the task of gesture recognition typically involves a number of challenges, including the possible variability in the dynamics of gestures, change in environment, occlusion, and the scarcity of high-quality well-annotated data.

To improve robustness against environmental variations and noise, many HGR frameworks also adopt multi-modal approaches, specifically, by including keypoints as one of the input streams alongside image input [2, 3].

Recent advancements in deep learning models and frameworks for keypoint extraction, such as MediaPipe [4] and DWPose [5], have also set higher and higher expectations for gesture recognition systems, from recognizing limited sets of static gestures, to also distinguishing within a large variety of dynamic gestures often with only subtle differences.

1.2 Challenges

In this project, we mainly rely on extracted keypoint landmarks instead of raw image frames for hand gesture recognition. While this approach reduces computational complexity, it remains challenging and non-trivial to discriminate between gestures that only differ subtly, such as small variations in the amount of bend of a finger.

For example, naive distance-based methods, like simple nearest-neighbour classifier on raw landmark coordinates, often misclassify gestures with high intra-class variations. For example, "open hand" gestures (e.g. G8 in Figure 1) can exhibit significant variation in finger positions, leading to accumulated shifts in phase space comparable to those between two distinct signs (e.g. Signs G4 and G8 in Figure 1).

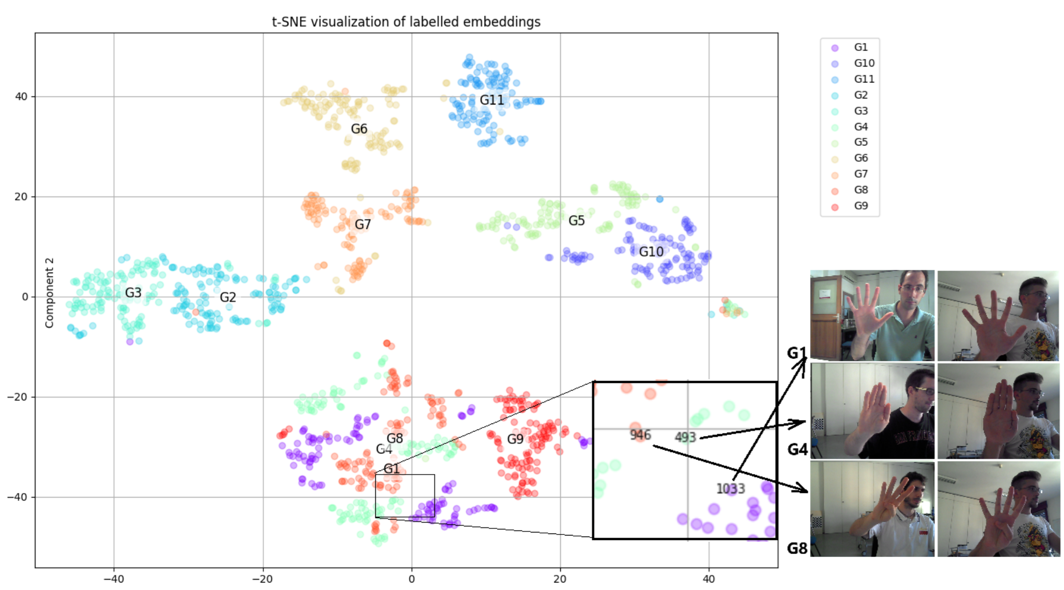

Figure 1: t-SNE plot for raw landmark over the Senz3d dataset

Such shifts in the phase space can also result from slight rotations or perturbations of individual joints, which can bring distinct hand configurations to close neighbourhoods in the feature space. Furthermore, inconsistencies and inaccuracy in depth estimation can introduce extra noise that further complicate discrimination between gestures.

1.3 Motivation

While existing deep learning approaches that utilize keypoint landmarks have achieved excellent performance in hand gesture recognition tasks [1], many existing frameworks are trained over fixed sets of gestures, which make them less suitable for applications requiring user-defined custom gestures, or for cases when high-quality annotated datasets are unavailable or too expensive to obtain. Consequently, traditional frameworks may generalize poorly to gesture variations that appear infrequently within the training set, in particular, performance over uncommon orientations or poses may degrade significantly.

To address this issue, semi-supervised contrastive learning emerges as a promising strategy with its ability to leverage limited sets of annotated data alongside much larger collections of unlabelled samples which are easier to obtain.

At the same time, the development of methodologies for parametrized

3-dimensional hand shape generation, such as MANO [6], has enabled the efficient generation of diverse and realistic gestures by sampling from the parameter space. This allows controlled generation of numerous similar hand configurations, which can be combined with contrastive learning approaches to allow training on a significantly wider range of orientations and poses than typically available in limited real datasets, potentially enhancing generalization to uncommon variations or poses.

1.4 Scope and Application

In this project, we aim to develop a unified, semi-supervised contrastive-learning framework for hand gesture recognition that enables quick adaptation to various downstream tasks, such as human-computer interaction and sign language, with minimal re-training or fine-tuning if possible.

Throughout the project, we will explore the effectiveness of various designs that we hypothesize would enhance the generalization capability of the models. Specifically, we have explored the following:

Static-Pose Representation Learning. We compared three static-input encoder architectures, Multi-layer Perceptron (MLP), Graph Convolutional Network (GCN) and Graph Attention Networks (GAT), which map hand landmark inputs of shape 21×3 into feature embeddings of size 128. These models are trained under various settings to test the effectiveness of different hypotheses:

1. **Graph-based models, capable of utilizing the edge information, would outperform MLP in terms of accuracy and the speed of convergence**. This is tested by training all three base models with supervised contrastive loss only on the Lexset dataset and evaluating the accuracy on the test set.

2. **Leveraging a large unlabelled dataset together with curriculum learning would improve generalization**. This is tested via training the models jointly with the synthetic MANO data as the unlabelled dataset while applying curriculum-based augmentation schedules.

Extension to Dynamic Gesture Recognition. After examining various approaches for extracting feature information from static input, we also aim to extend our contrastive approach to dynamic gesture recognition. To model temporal dependencies, we utilize sequential architectures such as RNN and LSTM. In this section, our investigation would focus on how contrastive learning approaches and augmentations transfer to dynamic tasks, in which the effectiveness is evaluated on the IPN dynamic gesture dataset.

2 Related Work

2.1 Contrastive Learning Approaches

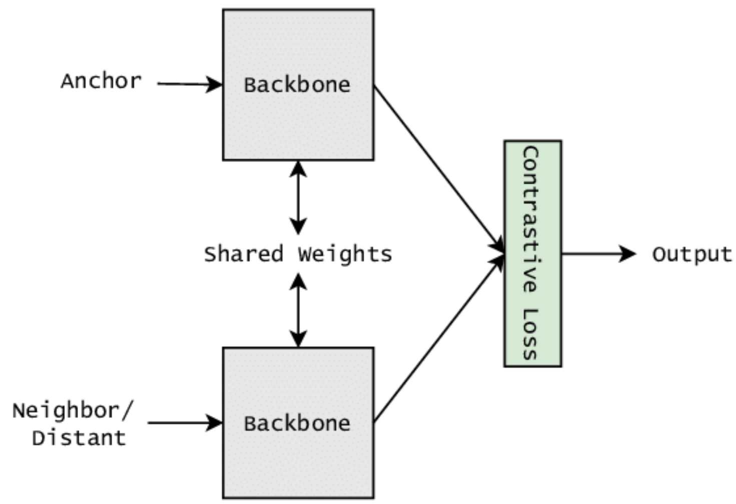

Figure 2: Architecture of a Siamese Network, with Contrastive Loss [7, 8].

In static image recognition, contrastive learning techniques, in particular, Siamese Networks [9], have played a crucial role under unsupervised and semi-supervised settings by utilizing shared parameters among the encoders for contrastive pairs, as illustrated in Figure 2, and have demonstrated its value in multiple fields including face identification. The architecture has contributed to the development of later frameworks such as SimCLR [10], MOCO [11], BYOL [12], etc., and have shown success on utilizing large sets of unannotated data by generating positive samples via data augmentation.

InfoNCE Loss. The InfoNCE loss is a well-known loss in the field that maximally utilizes the data within a batch of augmented duplets

$(x_i, x_i^+), i \in [1..B]$ with the latent $(z_i, z_i^+) = (f(x_i), f(x_i^+))$, with the following formulation: [13, 14, 15]

$$\mathcal{L}_{\text{InfoNCE}} = -\mathbb{E}_{i \in I} \left[\log \frac{\exp\left(\text{sim}(z_i, z_i^+)\right)}{\exp\left(\text{sim}(z_i, z_i^+)\right) + \sum_{j \in I \setminus \{i\}} \exp\left(\text{sim}(z_j, z_i^+)\right)} \right]$$

where $\text{sim}(x, y) = (x \cdot y)/\tau$ for temperature τ and $f(x) = \text{Proj}(\text{Enc}(x))$ for projector network $\text{Proj}(\cdot)$ and encoder network $\text{Enc}(\cdot)$.

For the contrastive models trained with InfoNCE loss, *cosine similarity* is instead used to compute the similarity between any two latent feature vectors, as it is optimized within the loss. While the InfoNCE loss is particularly effective in semi-supervised settings by contrasting augmented pairs (the ”positive samples”) against other instances (which are known as the ”negative samples”), it is adopted as the primary objective for training the unlabelled dataset in this project.

Supervised Contrastive Loss (SupCon). The supervised contrastive loss [16] utilizes also the information from the data from the same class apart from the augmentations, utilizing more comprehensively the information from the labelled datasets while preserving the properties of contrastive learning. It is given by:

$$\mathcal{L}_{\text{supcon}} = -\mathbb{E}_{i \in I} \mathbb{E}_{p \neq i : y(p) = y(i)} \left[\log \frac{\exp\left(\text{sim}(z_i, z_p)\right)}{\sum_{a \neq i} \exp\left(\text{sim}(z_i, z_a)\right)} \right]$$

where, the supervised contrastive loss utilizes all augmented versions of all positive samples with the same class $y(p) = y(i)$.

2.2 Keypoint-Based Approaches

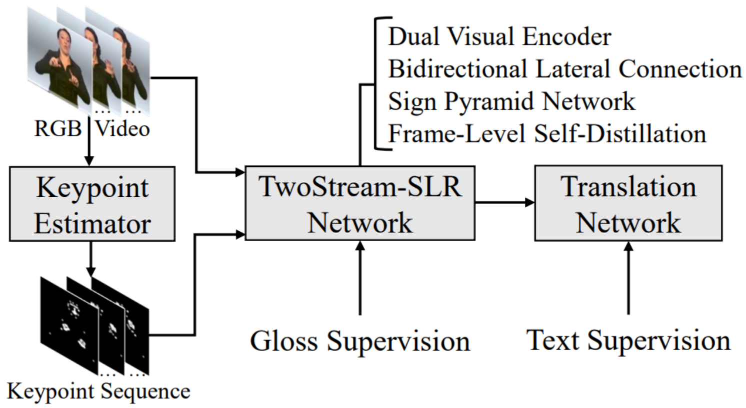

Figure 3: Architecture of TwoStreamNetwork [2].

With sophisticated pre-trained pipelines for extracting keypoints of hand and body gestures from videos, such as DWPose [5] and MediaPipe [4], many "skeleton-based" HGR models, like Sung et al. [17], TwoStreamSLR [2], and STGCN-GR [1], are integrating keypoints as a new input stream (See Image 3). This integration can theoretically improve robustness against environmental variations, given a reliable pre-trained keypoint extraction model like MediaPipe. Additionally, a wider variety of augmentations, such as translation and rotation of individual joints, are easier to implement compared to using the video stream directly.

2.3 Dynamic Gesture Recognition

While HGR models for static gesture recognition have become sophisticated, dynamic gesture recognition has remained a challenging task, especially over a large number of classes. Methods like TwoStreamNetwork [2] and SlowFastSign [18] perform convolution also across the time dimension to integrate temporal data and has still been in the leading position over the task of sign language recognition.

Contrastive Learning over Temporal Data. Techniques such as Contrastive Predictive Coding [15] and Temporal Contrastive Learning [19], e2eET [3] have been discussed in the literature, which utilize different parts of the same clip and different clips as the positive and negative samples.

2.4 Curriculum Learning

While augmentations are often beneficial and have been widely used for training models that are robust to perturbations, overly aggressive or complicated augmentations introduced too early may hinder the efficiency of training. Curriculum learning addresses this problem by training with increasing complexity and difficulty over time [20].

3 Static Gesture Recognition

In the first section of our work, we focus on improving the generalization ability of the backbone model with more comprehensive augmentations and utilization of large unlabelled datasets.

3.1 Methodology

3.1.1 Datasets

Supervised Datasets. Extending on previous work, we trained and evaluated our models on various static datasets, which is listed in Table 1. Specifically, the RWTH-PHOENIX-Weather 2014 MS Handshapes Dataset

Dataset (* indicates imbalanced dataset)	**# Classes**	**# Samples**	**Image Size**
Synthetic ASL Alphabet (Lexset) [21]	27	27,000	512×512
Senz3D [22]	11	1,320	640×480
RWTH-PHOENIX-Weather 2014 MS Handshapes [23]	60*	3,000	92×132

Table 1: Overview of the static hand-gesture datasets used in this project.

[23] (Figure 4) is a largely unbalanced dataset featuring around 3,000 images in total, with over 60 different hand shapes. In this project, we are utilizing the diversity of the hand shapes within this dataset.

Unlabelled Datasets. We also utilize the MANO hand model library [6] to generate a dataset of anatomically reasonable hand shapes by sampling and perturbing the low-dimensional parameter space (Figure 5).

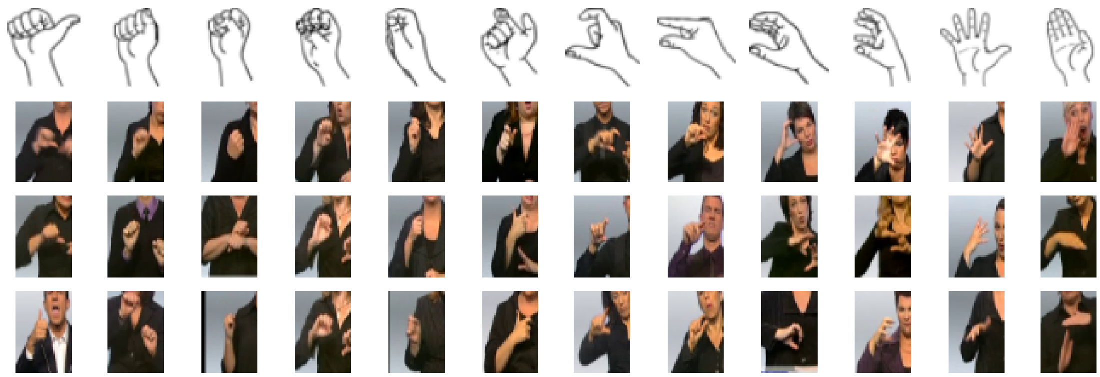

Figure 4: Sample images from the RWTH-PHOENIX-Weather 2014 MS Handshapes Dataset [23]

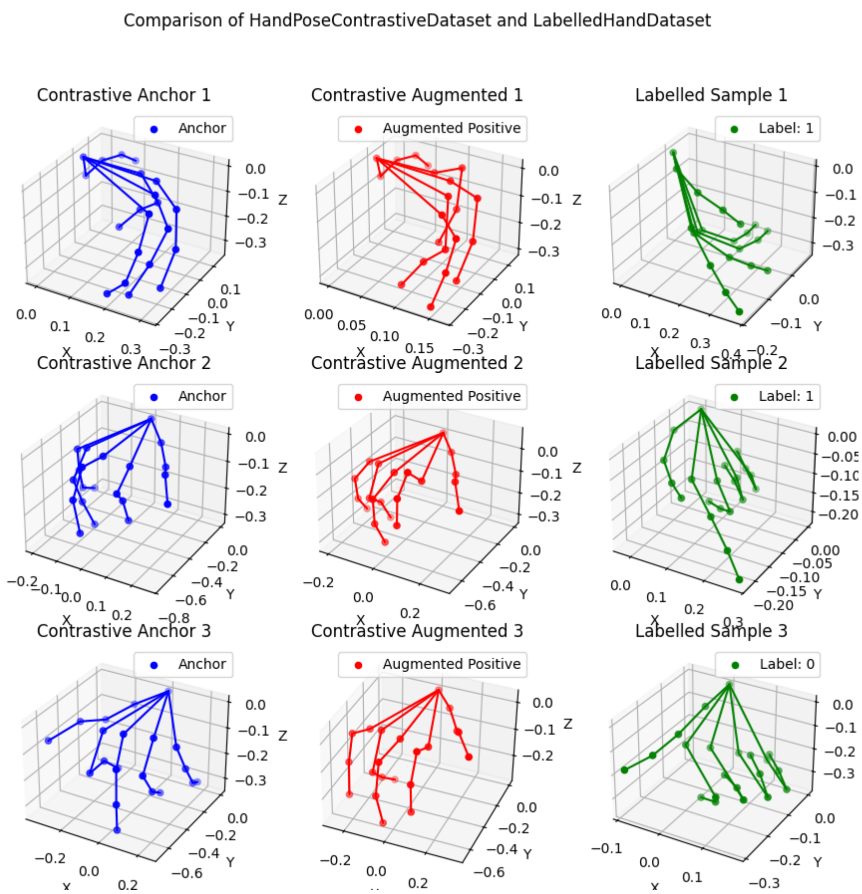

Figure 5: Comparison of MANO generated hands. Columns 1 and 2 show two augmented variants of the same base pose; Column 3 shows labelled samples from Lexset.

The generated dataset consists of $G = 32$ augmentations for $N = 100,000$ distinct base hand shapes, which is then used for semi-supervised portion of the training. The details of the mathematical formulation used for sampling in our implementation are provided in Appendix B.

3.1.2 Problem Formulation

The gesture recognition task can be defined as a function denoted by $G : \mathbb{R}^{21\times 3} \to \mathbb{R}^{C}$, where C is the number of classes within the dataset.

For a general gesture recognition model, we expect that there exists a model $M : \mathbb{R}^{21\times 3} \to \mathbb{R}^{D+H}$, where the first D dimensions within the output features is approximately invariant to the rotation or offsets of the hand, and the later H dimensions captures the remaining information. ie we can decompose M into $M = [M_{\text{pose}}, M_{\text{rot}}]$ where M_{pose} extracts the rotational-invariant pose of the gesture, and M_{rot} extracts the pose-invariant information of the gesture, like rotations.

In this report, we focus on the extractor M_{pose} for the rotationally invariant features of the gesture.

3.1.3 Model Architecture

For the static datasets, we performed preliminary tests using shallower models before proceeding to more complex architectures, as the latter require significantly longer training time.

We explored three major base encoder types, including Multi-Layer

Perceptron (MLP), Graph Convolutional Neural Network (GCN) [24] and Graph Attention Network (GAT) [25], where details will be covered in the next section.

We also explored the effects of input normalization (-I), utilization of angular parameterization (-A, or 6DoF) (as described in Appendix C) and pooling after the graph layers (-P).

3.1.4 Augmentations

For training the static-input extractors, we train over both the labelled and unlabelled datasets with increasing augmentation magnitudes across epochs, inspired by the idea of curriculum learning.

By allowing the model to learn first from easier samples, then progressively increasing the difficulty of the task (i.e. the magnitude of the augmentation), we hope to improve the training stability and allow for a more efficient convergence [20].

In our formulation, we gradually increase the maximum angle of rotation θ applied on the inputs, where the rotation matrix R can be decomposed into the base rotation matrix R_b which is consistent across the same batch, and the child rotation matrices R_a that are restricted to smaller angles but are applied individually to each augmented versions of the input. The details of the augmentation is provided in Appendix D.

Also, in the augmentation process, a total of $G = 4$ augmented versions of the same input entry would be created, extending the batch size from B to $B \times G$ and providing $G - 1$ positive samples for each sample.

3.1.5 Loss Functions

In the process of training static-input feature extractors, we adopt the InfoNCE loss for the *unlabelled* dataset $\mathcal{D}_{\text{unsup}}$ and the SupCon loss for the *supervised* dataset $\mathcal{D}_{\text{sup}}$. For each epoch, the model is trained over an equal amount of batches from each dataset (where $\mathcal{B}_{\text{sup}} \subseteq \mathcal{D}_{\text{sup}}, \mathcal{B}_{\text{unsup}} \subseteq \mathcal{D}_{\text{unsup}}$) and the final loss is given by:

$$\mathcal{L}_{\text{total}}(\mathcal{B}_{\text{sup}}, \mathcal{B}_{\text{unsup}}) = \mathcal{L}_{\text{SupCon}}(\mathcal{B}_{\text{sup}}) + \mathcal{L}_{\text{InfoNCE}}(\mathcal{B}_{\text{unsup}})$$

3.2 Experiments

3.2.1 Base Model Choice

For the selection of the optimal base encoder, we trained the three types of base encoders only with the supervised dataset Lexset using SupCon, and evaluated their performances over the test set of Lexset using k-Nearest Neighbour. We intentionally included the graph-based networks because hand landmarks naturally form a skeletal graph, where each node is connected to its neighbours via bones, which we believe that this relation can be exploited by graph-based networks for them to learn more expressive and meaningful representation of the hand poses. The architecture of the models are listed in Table 2:

Model	**Total Layers**	**Graph Layers**	**MLP Head** (#Layers, Hidden Size)
MLP	4	N/A	4 layers, [256, 128, 64]
GCN	5	3 graph convolutional layers	2 layers, [64]
GAT	5	3 graph attention layers (4 heads)	2 layers, [64]
Embedding Size	128	**Dropout Rate**	0.3
Activation	Leaky ReLU (slope=-0.01)	**Batch Norm**	After every layer

Table 2: Overview of Model Architectures for Static Hand Gesture Feature Extraction

Method (%) (* implies evaluation with Cosine Similarity); Evaluated with k-NN	**Accuracy**	**F1 Score**	**Convergence**	**Notes**
Raw Landmarks	96.69	96.69	-	Baseline embedding
6DoF	93.22	93.20	-	Angular parameterized
MLP*	97.39	97.37	540ep	
GAT-A (6DoF)*	*98.04*	*98.04*	200ep	No pooling
GCN-A (6DoF)*	**98.45**	**98.45**	200ep	No pooling

Table 3: Evaluation Results using K-NN and Learned Embeddings

Among the three base encoders, Table 3 shows that GCN and GAT demonstrate better performance and faster convergence without signs of overfitting. Hence, the following experiments will mainly focus on these two models.

3.2.2 Curriculum Learning

To prevent strong data augmentations from hindering the initial training of the model, we implemented a curriculum learning strategy as described in Appendix D, and obtained the following results:

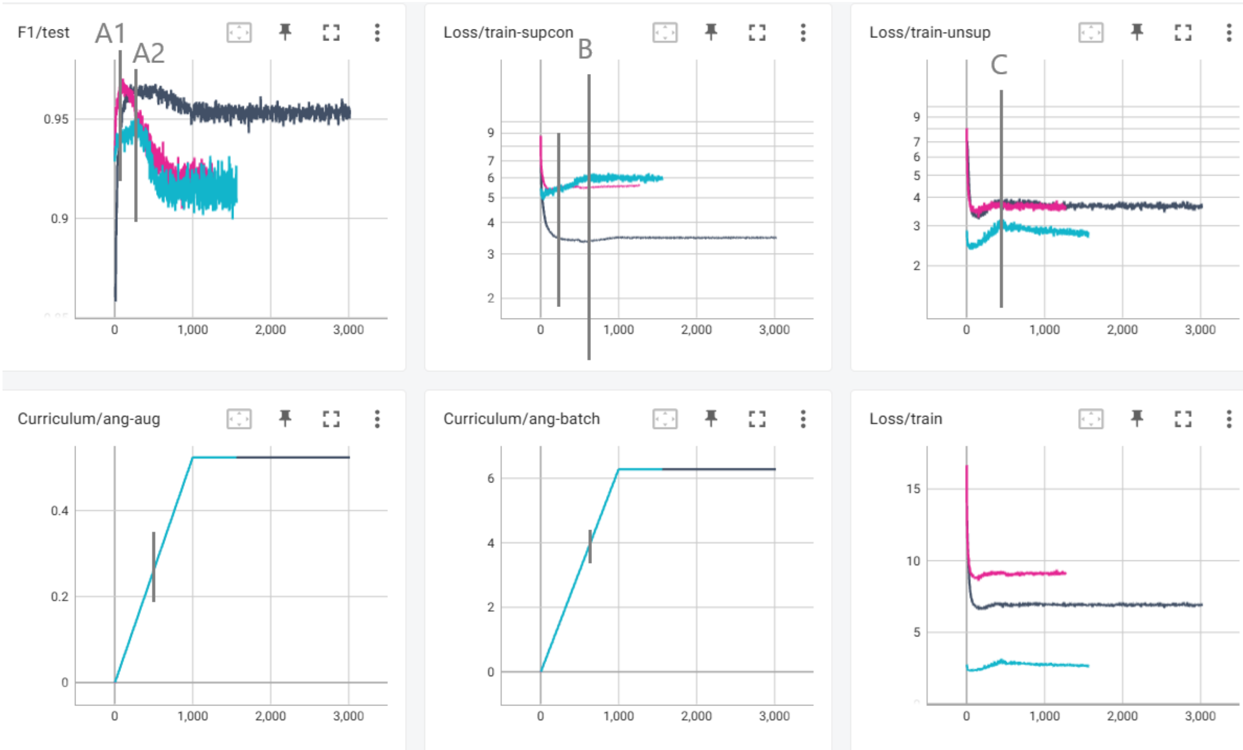

Figure 6: F1 score (top left), supervised (top middle) and semi-supervised loss (top right) for GAT (grey, pink) and GCN (blue) under curriculum learning, and their maximum augmentation angles (bottom left and middle)

As shown in Figure 6, curriculum learning leads to an initial increase in accuracy and F1-score, reaching a peak before declining and leveling off around a fixed value. In contrast, the accuracies never exceeded 90% if the augmentations are fixed to its maximum value throughout the training.

Contrary to our hypothesis, our experiments indicate that the models may have insufficient capacity to effectively learn when the augmentation magnitudes are overly large. Specifically, the accuracies peaked at an intra-batch augmentation magnitude of approximately 0.3 and batch augmentation magnitude of 4.

Furthermore, while the supervised loss increases beyond point B and stablizes, the semi-supervised loss continued to decrease after reaching a local peak at point C. This indicates that learning from the unlabelled dataset dominates when the augmentation magnitude is overly large.

Method (6DoF)	**Peak F1 (%)**	**Final F1 (%)**	**Handshape**	**Senz3d**
			Interclass (Mean ± Std)	
No Model	96.69		0.829 ± 0.111	0.917 ± 0.038
		Intraclass:	0.888 ± 0.085	0.955 ± 0.015
SupCon only, No Curriculum Scheduling				
GAT-A	97.63		0.996 ± 0.003	0.993 ± 0.004
		Intraclass:	0.998 ± 0.001	0.999 ± 0.001
GCN-A	**98.21**		0.997 ± 0.002	0.994 ± 0.003
		Intraclass:	0.999 ± 0.001	0.999 ± 0.000
With Unlabelled Dataset and Curriculum Scheduling				
GAT-A (pink)	**97.04**	91.44	0.721 ± 0.165	0.471 ± 0.261
		Intraclass:	0.903 ± 0.039	0.938 ± 0.026
GAT-A (grey)	*96.69*	95.46	0.824 ± 0.094	0.651 ± 0.156
		Intraclass:	0.930 ± 0.035	0.956 ± 0.020
GCN-A (blue)	94.88	90.81	0.216 ± 0.421	0.043 ± 0.461
		Intraclass:	0.747 ± 0.124	0.872 ± 0.038

Table 4: Comparison of peak and final F1 scores along with interclass statistics for curriculum learning runs. For the interclass and intraclass statistics, they are based on the distribution of the cosine similarities, where 1 indicates complete match and 0 indicates complete dissimilarity.

Table 4 and Figure 7 also showed that the clusters formed within the latent space managed to improve from the baseline for the GAT models, characterized by larger separations between the class means and lower intra-class variance, despite there being a decrease in the peak F1 score.

The result have demonstrated that training jointly with a large unlabelled dataset together with curriculum learning improves the ability for the models

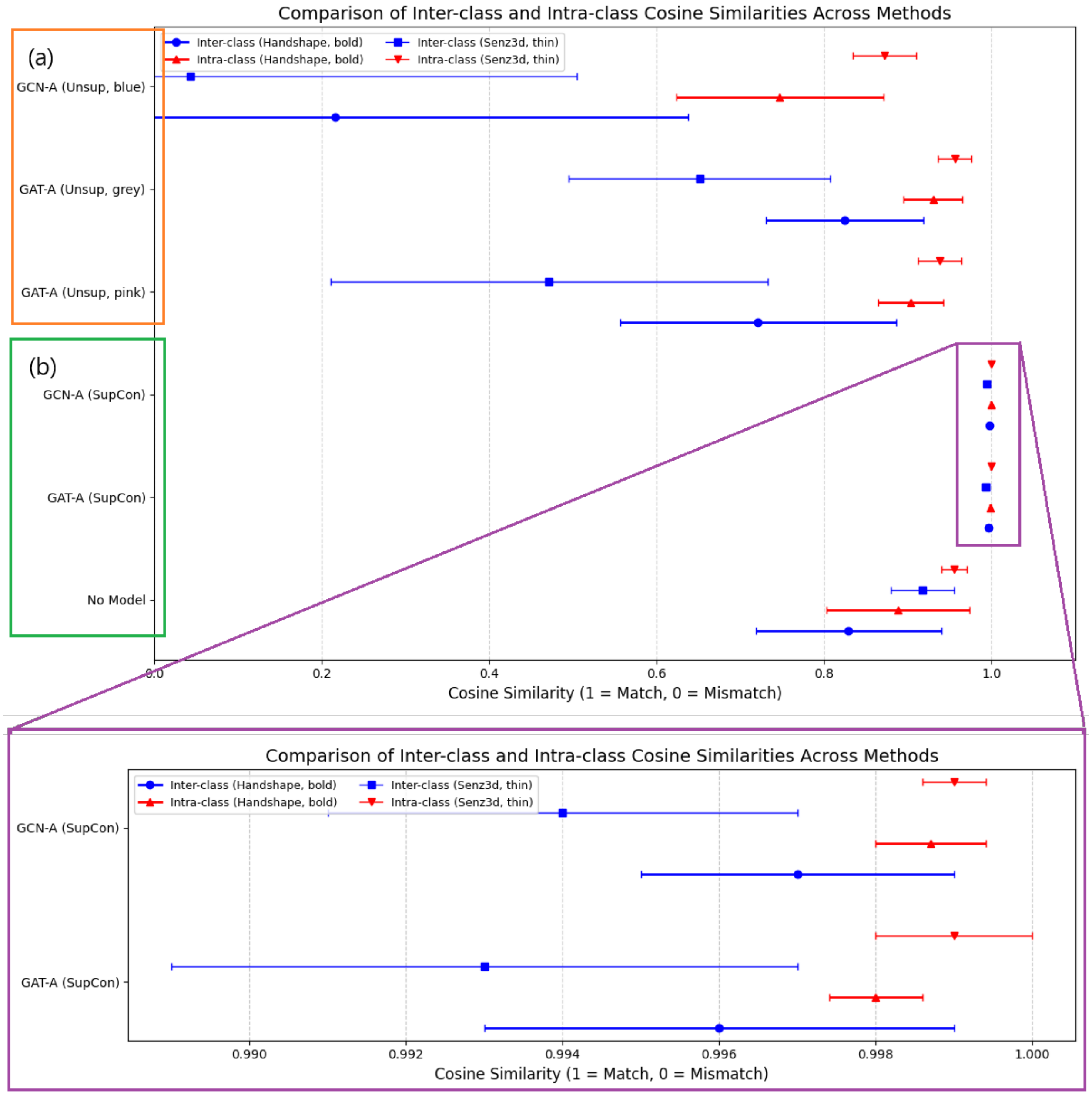

Figure 7: Comparison of interclass and intraclass statistics for models trained (a) with, and (b) without curriculum learning and unlabelled dataset.

to generalize to unseen datasets, despite the supervised training dataset is limited to only 26 gestures, while the evaluation dataset contains a broader set of 60 classes.

4 Dynamic Gesture Recognition

While we have tested the effectiveness of various approaches for extracting feature information from static input in the previous section, we also aim to extend our contrastive approach to dynamic gesture recognition in this work.

4.1 Methodology

4.1.1 Datasets

For the dynamic gesture recognition task, we primarily trained and evaluated over the IPN Dataset [26], which is a dynamic gesture dataset consisting of over 4,000 gesture instances in total, with 13 static and dynamic gestures for interaction with touchless screens (Refer to Figure 8 for samples). The full list of datasets used is listed as follows:

Dataset	**# Classes**	**# Samples**	**Image Size**
IPN Hand [26]	13	4,000 sequences	320×240
20bn-Jester [27, 28]	27	6,981 sequences	100×100

Table 5: Overview of the dynamic hand-gesture datasets used in this project.

4.1.2 Augmentations

For training dynamic gesture recognition models, we adopted the traditional approach of fixed augmentation, where each sequence of dynamic gestures is rotated about an arbitrary principle axis by an angle $\theta \in [-5°, 5°]$, and scaled along each axis by a factor $r \in [0.8, 1.2]$. Additionally, each sequence is also

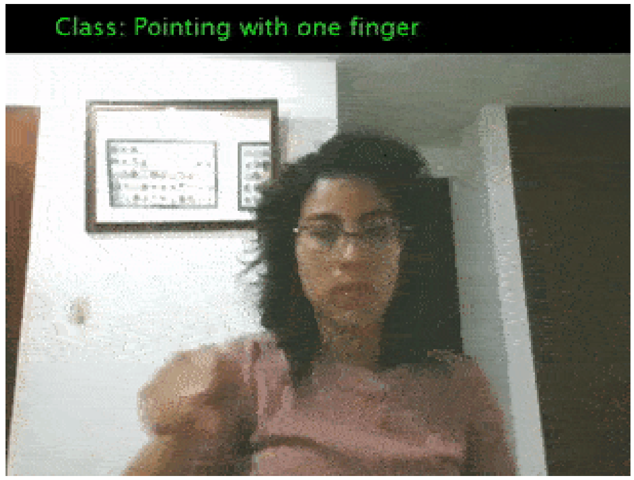

Figure 8: Sample images from the IPN hand dataset

linearly interpolated and sampled with random speed in the range 0.75x and 1.5x.

For models that rely only on hand keypoints (including LSTM-b, RNN, and the LSTM block responsible for hand inputs of the LSTM-hierachical model), the input coordinates of each node are normalized by expressing in terms of the position relative to the wrist node, which ensures that the model learns the underlying pose information that is independent of the global hand position.

4.1.3 Model Architecture

In this work, several architectures have been explored for capturing feature information from dynamic gesture inputs, as detailed in Table 6. The models are grouped into two categories, separated by the horizontal line in the table. Below the line, the LSTM-b and RNN models generate a single output feature

Model	Input Dim	# LSTM Layers	Hidden Dim	Pooling
LSTM-a	177	3	256	None
LSTM-windowed	177	3	256	AvgPool
LSTM-hierarchical	177	2/2/2 (body/hand/final)	256	None
LSTM-b†	63	2		
RNN†	63	2 (RNN)		

Table 6: Architectural details of the LSTM-based gesture models.
† Indicates that only hand keypoints are used; Otherwise, full-body keypoints are used.

embedding, based on an entire block of video, by applying an MLP classifier to only the final hidden state of the recurrent network.

In contrast, the models above the line perform frame-wise classification by applying the MLP classifier to the LSTM output at each time step. Meanwhile, the hierachical model adopts a part-wise architecture inspired by Song et al. [29], which separate LSTM branches processing body and hand landmarks. The resulting features from the branches are then concatenated and passed through a final LSTM block to output the frame-wise feature embeddings.

4.1.4 Contrastive Approaches

To facilitate a more efficient training with contrastive learning, we apply average pooling to the output using a window size of 32 frames, which allows the model to generate predictions for every window of 32 frames. In this report, models with this setup are marked with the suffix "-windowed".

Following the contrastive techniques from the previous section, the windowed models are trained using the SupCon loss, which is applied over the pooled features. Meanwhile, the two models that output a single prediction per video sequence are trained using a basic margin-based contrastive loss.

4.2 Experiments

4.2.1 Temporal-Based Contrastive Training

The results of our experiments evaluated on the IPN dataset, compared to the official benchmarks[26], are presented as follows:

The experimental results from Table 7 provide three major observations regarding the model architectures and the training approaches that are evaluated. First, comparing uni-directional and bi-directional LSTM variants, we observe consistently that bi-directional models achieve higher accuracy. For example, the Bi-LSTM-hierachical model achieved an accuracy of 84.78%, which leaded the equivalent uni-directional model (76.85%) by 7.93 percentage points.

Meanwhile, the results also demonstrate the effectiveness of using a hierachical or part-wise architecture over models that process full-body keypoints directly, with an increase of over 4 percentage points on average with this strategy.

Finally, we can also observe that models trained with contrastive methods yielded performance that is slightly lower compared to models trained directly

Model	**Input**	**Modality**	**Accuracy** (%)	**Convergence** (epochs)
Baseline Results reported by IPN dataset official website [26]				
ResNeXt-101	32 frames	RGB-Flow	42.47	—
ResNet-50	32 frames	RGB-Flow	39.47	—
ResNeXt-101	32 frames	RGB-Seg	39.01	—
ResNet-50	32 frames	RGB-Seg	33.27	—
ResNeXt-101	32 frames	RGB	25.34	—
ResNet-50	32 frames	RGB	19.78	—
Our Work (Direct Training)				
LSTM-a	framewise	Landmarks	71.13	300
Bi-LSTM-a^{2}	framewise	Landmarks	80.61	300
LSTM-hierarchical	framewise	Landmarks	76.85	600
Bi-LSTM-hierachical^{2}	framewise	Landmarks	**84.78**	400
Bi-LSTM-windowed^{*2}	32 frames	Landmarks	*81.72*	700
Bi-LSTM-w-h^{*234}	32 frames	Landmarks	**85.28**	800
Our Work (Contrastive)				
Bi-LSTM-windowed^{*2} (SupCon)	32 frames	Landmarks	*83.75*	700
Bi-LSTM-w-h^{*234} (SupCon)	32 frames	Landmarks	*84.95*	1000
LSTM-b (margin contrastive)	blockwise	Landmarks†	**94.02**	
RNN (margin contrastive)	blockwise	Landmarks†	76.81	
Our Work (Direct Application of Static-Trained Model)				
Temporal Similarity, Contrastive-K	blockwise	Landmarks†	72.34	

Table 7: Comparison of various architectures evaluated upon the IPN dataset.
† Indicates that only hand keypoints are used; Otherwise, full-body keypoints are used.
* With temporal augmentation 0.5x-5x speed.
^{2} Bi-LSTM indicates Bi-directional LSTM.
^{3} -w indicates that the inputs are cut into portions of 32 frames before prediction.
^{4} -h indicates that hierachical model is used.

for classification. With limited utilization of the unlabelled data and previous approaches like curriculum learning, this outcome suggests that either the true potentials of our contrastive method have not been fully utilized, or that our method does not have significant improvement over existing methods.

4.2.2 Direct Application of Static-Trained Models - Trail Analysis

Furthermore, by directly applying the model trained for static images, we can obtain a trail of latent representations of the dynamic gesture. The similarity between any 2 samples are then compared using metrics including DTW, average cosine distance and frechet distance. In most of the metrics, samples from the same class has mean less than the mean from different classes.

In this experiment, we utilized the Contrastive-K model from the previous semester as the base model. Table 8 shows the distributions of the intra-class and inter-class metrics obtained by evaluating over the 20bn-jester dataset.

Metric	Intra-class Mean (normalized)	Std	Inter-class Mean	Inter Std
DTW	1.0000	0.3649	1.2372	0.3416
Frechet	1.0000	0.3282	1.1905	0.3131
Avg. Euclidean	1.0000	0.4956	1.2082	0.4828
Avg. Cosine	1.0000	1.1631	1.3047	1.1988
Hausdorff	1.0000	0.3581	1.2064	0.3482
Correlation	1.0000	0.6871	1.4038	0.7355
Procrustes	1.0000	0.4392	1.1491	0.4077

Table 8: Intra-class and Inter-class Similarity Metrics for Direct Application of Static Model (raw)

For the classification task, we trained an extra 3-layer MLP with sigmoid activation that takes the above metric scores as input to predict if any 2 samples belong to same class. The accuracy obtained is 72.34%, which is significantly lower compared to other temporal models (Table 7). This result suggests that our previous contrastive model, with minimal training over the large unlabelled corpus of hand gestures, failed to generalize to unseen dynamic datasets.

5 Discussions

In this report, while we have covered various contrastive approaches over both static and dynamic gestures, the ambitious original objective of building a general, unified, and invariant hand gesture representation model remains only partially fulfilled.

However, our experiments provided key insights into the possible effective approaches. From our experiments on static gesture recognition, we observed that (1) graph-based networks are more effective and lead to faster convergence by utilizing edge connections that formed naturally from the skeletal structure of the hand, and (2) training jointly with a large unlabelled dataset together with curriculum learning improves the ability for the models to generalize to unseen datasets.

Our experiments on extending the contrastive approach to dynamic gesture recognition tasks have demonstrated the possibilities of (1) hierachical and part-wise architecture in leading to more sophisticated understanding of the inherent structure of the hand gestures, and that (2) that our attempt of incorporation of contrastive learning displayed no significant improvement over existing methods, which may hint the need of further investigation over more complex contrastive learning approaches.

5.1 Future Work

In the future, we expect later work can expand upon our work and to train a true, general hand gesture encoder that is able to capture the general rotation-invariant and scale-invariant information of gestures that can be used to quickly adapt to various downstream tasks including dynamic gesture recognition.

In light of the current work, we hope that later work could explore the possibilities of incorporating both static and dynamic datasets for joint training via curriculum and contrastive learning.

A Division of Labour

To ensure clarity in our contributions, Table 9 outlines the division of labor between group members for key project tasks.

Task	**TAM, Ka Ho**	**MAN, Tze Wa**
Planning	Ideation and Planning	-
Dataset Preparation	Generated and preprocessed unlabelled dataset using MANO hand model	Curated and preprocessed supervised datasets (Lexset, Senz3D, RWTH-PHOENIX, IPN Hand)
Curriculum Learning	Designed and implemented curriculum-based augmentation strategy for static models	-
Static-Input Models	Developed and tested MLP and GAT models, including input normalization and angular parameterization	-
Dynamic-Input Models	Designed and tested Bi-LSTM with SupCon and windowed approaches	Implemented RNN and margin contrastive LSTM models
Contrastive Learning	Configured semi-supervised contrastive augmentations (InfoNCE, SupCon) for static models	Developed contrastive learning for sequences (margin contrastive) and DTW integration
Experimentation	Conducted experiments for static models (MLP, GCN, GAT) and analyzed F1 scores	Ran experiments for RNN and LSTM models, evaluated loss trends
Demo	-	Implemented the demo program for demonstration.

Table 9: Division of Labor for Project Contributions

B MANO parameterization

In the parameter space for the MANO hand model, let

$$\beta_{\text{base}} \in \mathbb{R}^{10}, \quad p_{\text{base}} \in \mathbb{R}^{45}$$

denote the MANO perturbation and pose parameters. Define the index sets S_f, S_r and S_t which correspond to the list of indices of features controlling the forward bending, sideways bending (rotation) of each finger, and the bending of the thumb, respectively.

Then, we sample the base parameters for each entry as

$$\beta_{\text{base}} \sim \text{Uniform}(-\sigma_\beta,\, \sigma_\beta),\;\; p_{\text{base},i} \sim \mathcal{N}(-b_i,\; \sigma_{p,i}^2),$$

where

$$\sigma_\beta = 2.0,\;\; \sigma_{p,i} = \begin{cases} 2.0, & i \in S_f \cup S_t, \\ 0.3, & i \in S_r. \end{cases},\;\; b_i = \begin{cases} 1.5, & i \in S_t, \\ 0.5, & i \in S_f, \\ 0, & i \in S_r. \end{cases}$$

We generate G augmented samples from the base parameters by adding Gaussian noise:

$$\beta_{\text{aug}}^{(j)} = \beta_{\text{base}} + \varepsilon_\beta, \quad \varepsilon_\beta^{(j)} \sim \mathcal{N}(0,\, \tilde{\sigma}_\beta^2), \quad \tilde{\sigma}_\beta = 0.1,$$

$$p_{\mathrm{aug},i}^{(j)} = p_{\mathrm{base},i} + \varepsilon_{p,i}, \quad \varepsilon_{p,i}^{(j)} \sim \mathcal{N}(0,\, \tilde{\sigma}_{p,i}^2), \quad \tilde{\sigma}_{p,i} = 0.1\,\sigma_{p,i}.$$

Afterwards, the j-th generated hand shape is obtained via the MANO forward model

$$H^{(j)} = \mathcal{M}(\beta_{\mathrm{aug}}^{(j)}, p_{\mathrm{aug}}^{(j)}),$$

and is then normalized and rescaled so that the sampled hand shapes have coordinates approximately lie within the range $[-0.4, 0.4]$ and have the standard deviation of each coordinate axis $\sigma_x = \sigma_y = \sigma_z = 0.1$.

The generated dataset is of the shape $[N, G, 21, 3]$, where we chose the number of entries to be $N = 100,000$ and the number of augmentations per entry to be $G = 32$.

C Angular Parametrization

Implemented with Python library `scipy`, the relative quaternion orientations are computed from the difference in orientation compared to the previous node. The wrist node orientation is computed based on the facing direction of the palm, while the other node orientations are computed from the direction of the edge connecting to that node, so that the normal vector always point in the direction that the finger curls towards (See Figure 9).

This approach is adopted in hope of improving the model's understanding of relative rotations. The quaternion approach is adopted as it is known to be

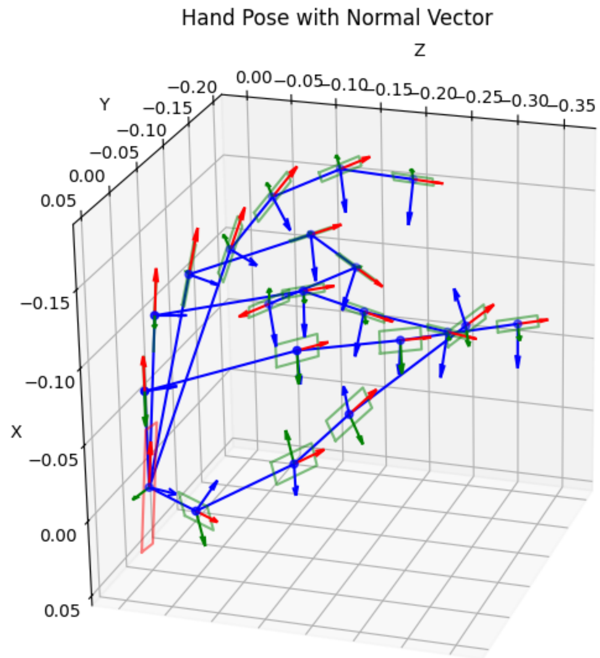

Figure 9: The axes corresponding to the "absolute orientation" of each hand landmark.

more robust in representing rotations while avoiding the typical Gimbal lock problem and inconsistency in rotation representation that would present in other methods like matrix representation or using Eurler angles [30].

D Curriculum Scheduling for Static Datasets

Given grid size G, we augment the input batch into the shape of $[B, G, 21, 3]$. That is, we would have G augmentations for every entry within the batch. The augmentation of each batch is generated as follows:

- Base augmentation: $R_b(t) = \text{RandomRotation}(\alpha_b \min\{t/T_b, 1\})$
- Child augmentations: $R_a^{(j)}(t) = \text{RandomRotation}^{(j)}(\alpha_a \min\{t/T_a, 1\})$
- For each batch from the supervised dataset $\mathbf{L} \in \mathbb{R}^{B\times C}, \mathbf{B} \in \mathbb{R}^{B\times 21\times 3}$,

the augmented sample have the shape $\mathbf{B}' \in \mathbb{R}^{B\times G\times 21\times 3}$, which is then flattened into the shape $(B \cdot G) \times 21 \times 3$.

  - Where, the j-th augmented sample of the i-th entry is given by: $\mathbf{B'}_{ij} = R_b(t)R_a^{(j)}(t)\mathbf{B}_i$

- For each batch $\mathbf{B} \in \mathbb{R}^{B\times G_D\times 21\times 3}$ from the unlabelled dataset that is generated with G_D augmentations, the augmented samples have the shape $\mathbf{B}' \in \mathbb{R}^{B\times G\times 21\times 3}$, which is then flattened into the shape $(B \cdot G) \times 21 \times 3$.

  - Where, the j-th augmented sample of the i-th entry is given by: $\mathbf{B'}_{ij} = R_b(t)R_a^{(j)}(t)\mathbf{B}_{iS_j}$ and $S \in \mathbb{R}^G$ is a random sample of size G from the range $[0, G_D)$.

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